

Integrals – 4

Series and integrals

Homework

1 сентября 2026 года

Theoretical minimum. Termwise integration and its justification (uniform convergence, monotone and dominated convergence); the expansions of $\frac{1}{1\pm x}$, $\ln(1+x)$, $\arctan x$; the formula $\int_0^1 x^a \ln^n x \, dx$; the identity $\Gamma(s)\zeta(s)$; Catalan's constant $G = \sum \frac{(-1)^k}{(2k+1)^2}$ and the dilogarithm $\text{Li}_2(z) = \sum \frac{z^n}{n^2}$.

1. Compute $\int_0^1 \frac{\ln x}{1-x} dx$, $\int_0^1 \frac{\ln x}{1+x} dx$ and $\int_0^1 \frac{\ln(1+x)}{x} dx$; explain where the factor $\frac{1}{2}$ in the last answer comes from.
2. Prove that $\int_0^1 \frac{\arctan x}{x} dx = G$, Catalan's constant, and that $\int_0^{\pi/4} \ln(\tan x) dx = -G$.
3. Compute $\int_0^1 \frac{x^n}{1+x} dx$ in closed form and deduce the asymptotics $\int_0^1 \frac{x^n}{1+x} dx \sim \frac{1}{2n}$ as $n \rightarrow \infty$.
4. Compute $\int_0^\infty \frac{\ln x}{1+x^2} dx$ and $\int_0^\infty \frac{x^{a-1}}{1+x} dx$ by expanding into a series in x on $(0, 1)$ and in $1/x$ on $(1, \infty)$.
5. Prove the identity $\sum_{n \geq 1} \frac{1}{n^2 \binom{2n}{n}} = \frac{\pi^2}{18}$ by writing $\frac{1}{\binom{2n}{n}}$ as a beta integral.
6. Compute $\int_0^1 \{1/x\} dx$, where $\{\cdot\}$ denotes the fractional part; Euler's constant γ should appear in the answer.
7. (Integration Bee 2020) Compute $\int_0^1 \ln \frac{1+x}{1-x} dx$, and also $\int \frac{1+e^x}{1-e^x} dx$ (2015); compare where a series expansion helps and where it gets in the way.
8. Prove the identity $\int_0^1 x^{-x} dx = \sum_{n \geq 1} n^{-n}$ and its alternating counterpart, estimating the remainder and stating how many terms are needed for accuracy 10^{-6} .

9. Compute $\int_0^\infty \frac{\sin x}{e^x - 1} dx$ by expanding $\frac{1}{e^x - 1}$ into a geometric series, and express the answer through the cotangent.

10. Prove the Abel–Plana formula for rapidly decreasing f :

$$\sum_{n \geq 0} f(n) = \int_0^\infty f(x) dx + \frac{f(0)}{2} + i \int_0^\infty \frac{f(it) - f(-it)}{e^{2\pi t} - 1} dt,$$

and verify it on $f(x) = \frac{1}{(x+a)^2}$.